Fine-Grained Bird Species Classification Using EfficientNet-B5 and Transfer Learning

The data was split into **training, validation** and test sets to precisely evaluate it. Preprocessing of the images was done by resizing, normalisation and data augmentation (**random resized crop, horizontal flip, rotation and colour jitter)** to alleviate **overfitting** and enhance generalisation. The **EfficientNet-B3** (pre-trained on ImageNet) model was used, and the final classification layer was altered to the target classes to allow **transfer learning** and converge faster.
The model has been trained with a **batch size of 16**, **CrossEntropyLoss** and **AdamW optimizer** with the learning rate of 0.001 and a weight decay of 1e-4. Such hyper-parameters were selected according to the general practice of transfer learning and minor project-specific experiments to ensure stable validation results (as opposed to hyper-parameter optimization). The model was trained to 70 epochs, with the highest **validation accuracy of 86.50%**. Random seeds were frozen and the model checkpoints were stored to make it reproduce.
Final test set evaluation resulted in an **Accuracy of 85.21, Precision of 0.8561, Recall of 0.8533** and **F1-score of 0.8508**. Accuracy is used to monitor the percentage of correct predictions. Precision refers to the number of correct predicted positive samples and lowers false positives. Recall is used to minimize false negatives and identify how many true positive samples were actually detected. **F1-score** is a single integrated measure of precision and recall and is thus applicable in situations where it is necessary to measure the general performance of the classification. **A confusion matrix** was as well constructed to examine the errors of the classes in prediction.

Models Hyperparameters:

Model	EffiecientNet-B3
Optimizer	AdamW
Loss Function	Cross-Entropy Loss(0.05)
Learning Rate	1e-4
Weight Delay	1e-4

Batch Size	16
Epochs	70
Scheduler	ConsineAnnealingLR(50)
Image Size	380*380
Mixed Precision Train	Enable(AMP GradScaler)

The model which was used is **EfficientNet-B3** with the **AdamW optimizer** and **Cross-Entropy Loss (label smoothing = 0.05)**. Training was performed using a **learning rate of 1e-4**, **weight decay of 1e-4**, **batch size 16**, and **70 epochs**, with a **CosineAnnealingLR scheduler (T_max = 50)** to improve convergence.
Images were resized to **380×380**, and **mixed precision training (AMP GradScaler)** was enabled to reduce memory usage and speed up training while maintaining performance.

Results: Test Accuracy

Accuracy	**Precision (macro)**	**Recall (macro):**	**F1 Score (macro):**
85.21	0.8561	0.8533	0.8508

Model 2:

Model 2 which was developed is self-designed CNN architecture, where convolutional layers, activation functions, pooling layers, and fully connected layers were manually defined instead of using a pre-trained network. The model learns features directly from the dataset rather than relying on transfer learning. This approach was used to compare the performance of a custom-built model against the pre-trained EfficientNet-B3 and to analyse differences in accuracy, generalisation, and training behaviour.

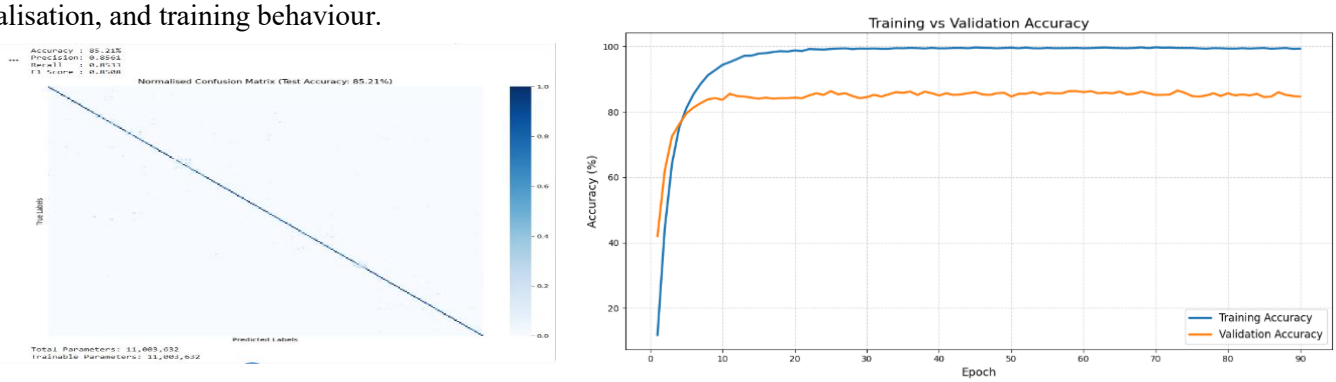
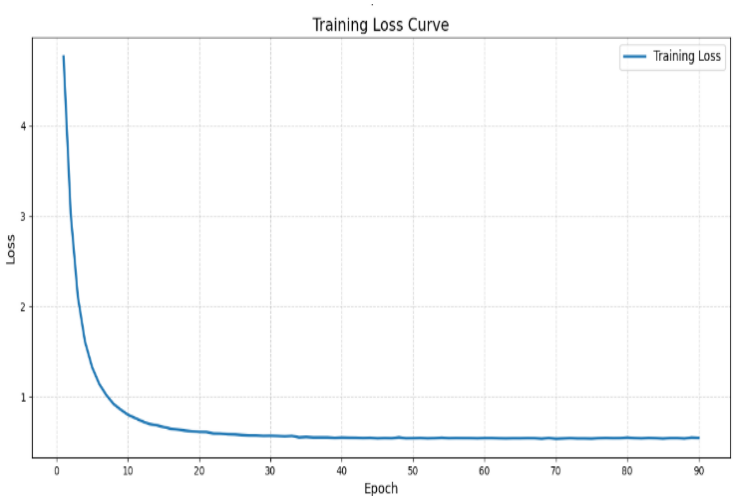


Fig :Plots of Confusion matrix with accuracy 85.21 and Model Accuracy



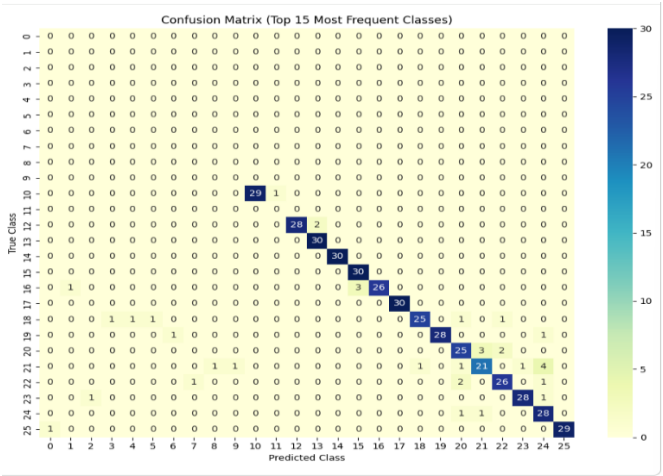


Figure:Training loss and Top 15 Frequent Classes

Fine-Grained Image Classification

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Generating example predictions visualization (Model 1)...
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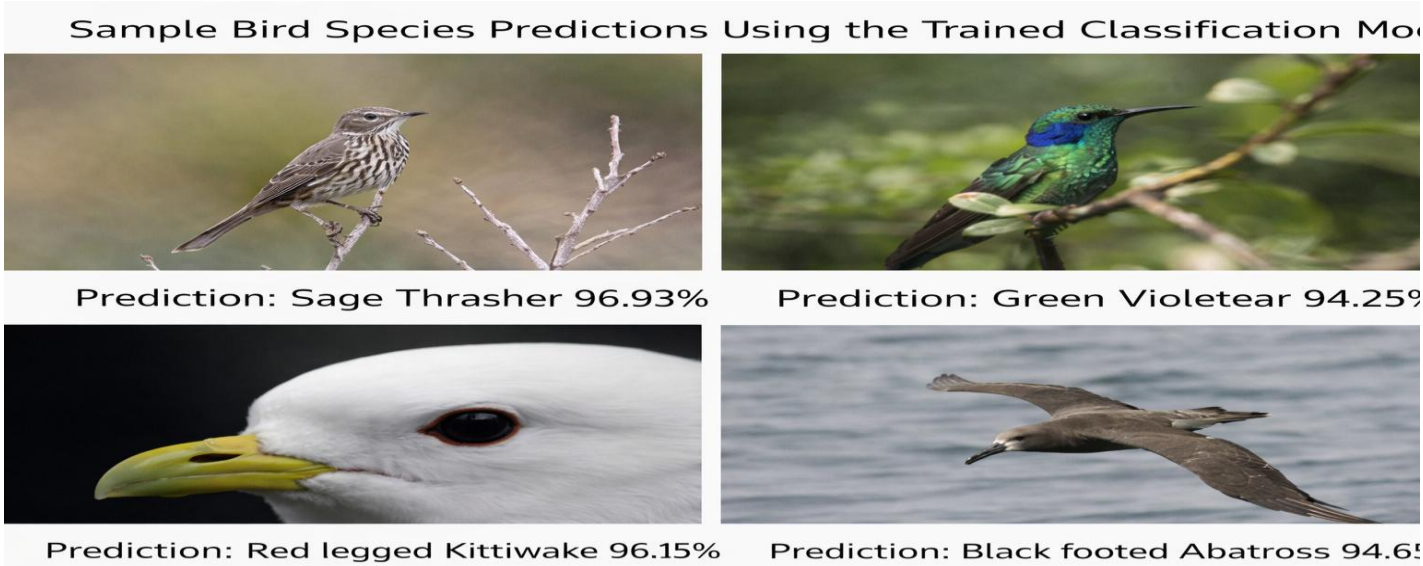
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Test Set Performance Summary (Model 1):
                Metric   Value
0             Accuracy  85.21%
1   Precision (macro)  0.8561
2      Recall (macro)  0.8533
3    F1 Score (macro)  0.8508
Saved: /content/drive/MyDrive/DataSet/DL_Dataset/CUB_200_2011/test_metrics_summary_model1.csv
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